Material Science 1: Deformation and failure group work

Group number: M4

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Introduction

In order to understand how the properties of different objects change under stress, we designed an experiment to measure the properties of metal spoons, cans and paper clips under stress. This poster will show you the experimental design of each sample, the corresponding experimental results and analysis conclusions.

Objective of Product

I. Experimentally investigate the mechanical properties of three different objects: elastic deformation, plastic deformation and failure.

II. Develop our analytical, problem-solving and collaborative skills by working together on experiments and poster designs.

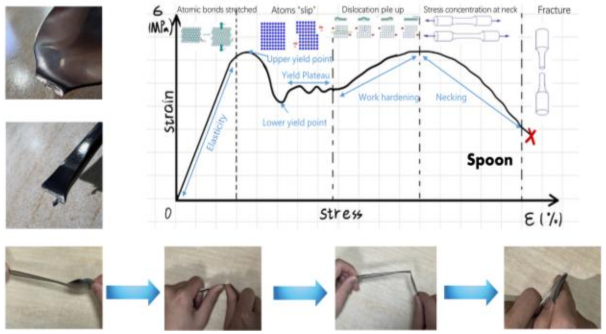
Methods of experiment

I. Place each of the three items horizontally on the table.

II. Holds the ends of the item with both hands and performs bending and twisting operations on it

III. Take photos during the deformation and failure of the items and draw the stress-strain diagrams by analyzing the magnitude of the applied force and the deformation.

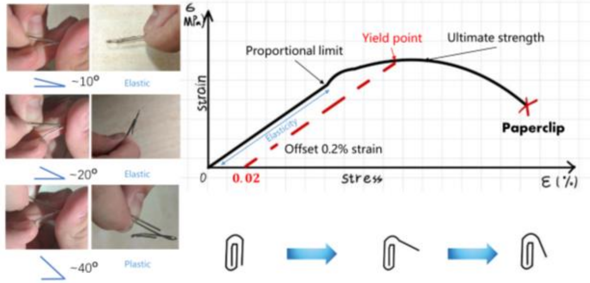
Results and Analysis

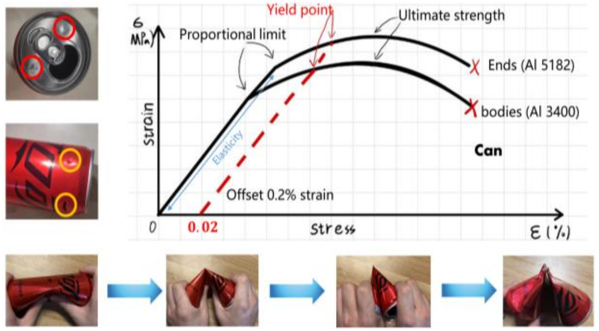


Spoons:
Ductile Failure
Martensitic stainless steel
Rich in chromium and carbon
good ductility, plasticity and hardness. [1]

We bend it so that it fatigues at the handle of the spoon. It fatigues and breaks at the point where the head and tail of the spoon meet. Macroscopically, the fracture were striated. The comparison results show that the deformation of the spoon handle is smaller under the same stress.

Paper clips:
Brittle failure
Made of iron with good resilience.
First, we gently bend a paper clip to a small angle and find that it returns to its original shape, due to its very good toughness. This material deforms before it fails because atoms readily rearrange themselves within the grain, and defects called dislocations promote this rearrangement. [2]

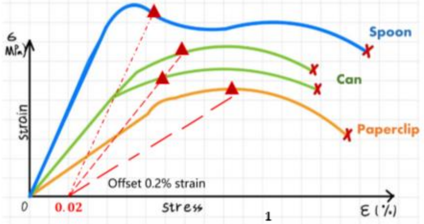




Can: Brittle failure Aluminum Alloy.
We first compared the deformation of the top and side surfaces of the can under the same stress, and concluded that the side surfaces were not as hard as the top surface, and that the deformation was more pronounced under the same stress. This is because its sides are usually made of aluminum 3400, while the top and bottom surfaces are made using the thicker and stronger aluminum 5182. Aluminum 5182 has a high degree of hardness.
Due to the presence of many small irregular cavities in aluminum alloys, the grains tend to slip and eventually become brittle. [4] Eventually it undergoes brittle failure and breaks open.

Conclusion

In the experiment, we combine the theoretical knowledges with daily life. All three of these items have been known to crack and eventually break, such as a spoon breaking at the joint, a paper clip breaking, and a can break in the middle. That's because, when the crack tip stress (σm) exceeds the critical stress (σc), the object cracks and undergoes brittle damage. (See Equation 1，2 and 3.) Also when the stress intensity factor reaches a value of Ka, an unstable fracture occurs. (see Equation 4). Paper clips and cans are brittle failures so after applying the cyclic stresses, fracture occurs.[3]

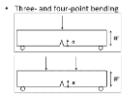
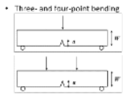
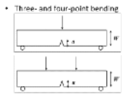


Equation 1: $\sigma_m = 2\sigma_0\left(\frac{a}{\rho_t}\right)^{\frac{1}{2}} = K_t\sigma_0$

Equation 2: $\sigma_c = \left(\frac{2E\gamma_s}{\pi a}\right)^{\frac{1}{2}}$

Equation 3: $\sigma_m > \sigma_c$

Equation 4: $K_{IC} = Y\sigma_c\sqrt{\pi a}$

Method	 soda can	 spoon	 paperclip
Deformation	 Three- and four-point bending	 Three- and four-point bending	 Three- and four-point bending
	Elastic deformation		
	Plastic deformation		
Failure	brittle	ductile	fatigue

Reference

[1] Connor, Nick. "What Is Martensitic Stainless Steel - Definition." Material Properties, 31 July 2020, material-properties.org/what-is-martensitic-stainless-steel-definition/. Accessed 29 Nov. 2023.
[2] Kruzic, J. J. "Predicting Fatigue Failures." Science, vol. 325, no. 5937, 9 July 2009, pp. 156–158, https://doi.org/10.1126/science.1173432. Accessed 18 May 2019.
[3]Ewing, James R, and J. C. W. Humfrey. "VI. The Fracture of Metals under Repeated Alternations of Stress." Philosophical Transactions of the Royal Society of London, vol. 200, no. 321-330, 1 Jan. 1903, pp. 241–250, https://doi.org/10.1098/rsta.1903.0006. Accessed 24 Apr. 2023.
[4] Kobayashi, T. "Strength and Fracture of Aluminum Alloys." Materials Science and Engineering: A, vol. 286, no. 2, 15 July 2000, pp. 333–341, www.sciencedirect.com/science/article/pii/S0921509300009357, https://doi.org/10.1016/S0921-5093(00)00935-7.

Materials Science 1
Deformation and Failure
Reflection of Group work

1. **Describe the focus of the reflection. What was the objective you set out to achieve? Did you achieve this objective?**

In the experiment we measured the properties of metal spoons, cans and paper clips under different stresses by applying several different forces to different materials. After that we analyzed the three items according to the concepts of deformation and failure fracture mechanism and through this experiment we achieved the desired purpose.

2. **Why did the team achieve/ not achieve your objectives? What challenges did you face? How did you overcome these challenges?**

The most important reason we achieved our goal was that we fully analyzed the experiment before it started, we considered two ways of stress deformation for each material, and everyone in the group was actively involved in the experiment to ensure that it went smoothly and efficiently. We encountered the challenge of not being able to measure the specific stresses applied to fail the three items, and then we were able to roughly determine the range of stresses applied by ranking the ease of failure of the items through the individual experimental perceptions of the group members.

3. **What have the team learned from this?**

One of the most important things that all group members learnt this time was to think critically, especially questioning each other during the brainstorming session, because in this assignment we found that 3 members' analyses of the properties of the metals were not correct when they were summarized at a later stage. Therefore, we re-questioned and discussed what we already had and finally came up with the correct conclusions. Due to the variety of measurement methods, we get different images and fatigue properties from different measurement methods, so sometimes there is a gap between theoretical knowledge and actual measurements.

4. **What changes will you make in the way you work as a team?**

Group course work went smoothly this project. Everyone does their own job and completes the assigned tasks with high quality.
For future improvements, this group work taught us to check each other's work, after all, sometimes we can only see the shortcomings and flaws of the article's thinking by thinking outside of ourselves.
At the same time, as there was no record of the person who co-ordinated the time this time, control of time still needs to be improved

5. **State the contributions from each member of the group to complete the experiment, discussion of results, poster and reflection.**

(Example: Student 1: Andy Bushby, designed poster, created graphs, wrote Q4 of reflection)
Student 1:Tianjian Feng, designed poster, made graphs, recorded the result of experiment
Student 2:Qiuyu Zhu, wrote objective of experiment, the method of the experiment and finished Q1&2 of reflection
Student 3: Yixuan Yuan, completed the analysis of findings, wrote conclusion
Student 4: Zeyin Jin, made graphs, accessed to articles, finished Q3&4 of reflection
Student 5: Tianche Sun, completed the experiment, wrote introduction part
Student 6: Fuxu Wang, Helped to complete the experiment